Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/13/2013
PI Name:	Keller, David Ph.D.		
Project Title:	Thermoregulatory Capacity and Exercise Responses following Prolonged Lunar Analog Bedrest		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	76019-0259	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	10/01/2008	**End Date:**	09/30/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Received extension to 9/30/2012, per C. Guidry/NSSC (Ed., 9/26/2011)		
Key Personnel Changes/Previous PI:	One graduate assistant has been added to the project (FY11 report).		
COI Name (Institution):	Crandall, Craig (Presbyterian Hospital of Dallas)		
Grant/Contract No.:	NNX08AY40G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Impaired exercise-thermoregulation and altered cardiovascular responses to exercise (i.e., reduced work capacity, elevated heart rate for a given exercise workload, attenuated maximal cardiac output, etc.) occur following actual and simulated microgravity exposure. It is unknown whether the thermoregulatory and cardiovascular responses to dynamic arm or dynamic leg exercise at the same metabolic demand (i.e., VO2) are comparable. Furthermore, the test, re-test reproducibility of these different exercises have not been investigated from an exercise-thermoregulatory standpoint. The studies proposed in this application will compare the exercise-thermoregulation and cardiovascular responses to prolonged, sub-maximal arm versus leg exercise. Findings from these studies will provide important information related to the usefulness and appropriateness of the use of different exercise protocols from both experimental and in-flight countermeasure perspective.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The benefit to life on Earth associated with this research is related to an improved understanding of body temperature responses to different types of exercise (i.e., Arm cycling and Leg cycling). These findings can serve as a foundation for the usefulness of, for example, arm cycling for heat acclimation purposes.
Task Progress:	Project Synopsis: The ability for the human body to maintain a certain body temperature during exercise is impaired following exposure to space (actual exposure and simulated microgravity, e.g., bed rest). In addition, cardiovascular responses to exercise are also impaired following this exposure (i.e., reduced work capacity, elevated heart rate for a given exercise workload, attenuated maximal cardiac output, etc.). It is unknown whether the “thermoregulatory” and cardiovascular responses to different types of exercise (e.g., arm or dynamic leg exercise) at the same intensity are comparable. Furthermore, the reproducibility of these different exercises has not been investigated from a thermoregulatory standpoint. The studies proposed in this application will compare the body temperature and cardiovascular responses to prolonged arm versus leg exercise. Findings from these studies will provide important information related to the usefulness and appropriateness of the use of different exercises from both experimental and in-flight countermeasure perspective. The studies from this grant will characterize thermoregulatory and cardiovascular responses to arm and leg cycle exercise. Additionally, the thermoregulatory and cardiovascular responses to arm and leg exercise will be examined on multiple days to examine potential day-to-day differences in the responses to exercise. Findings from this proposal are expected to have an important impact on the understanding of different types (i.e., “modes”) of exercise on exercise-thermoregulatory responses (as well as cardiovascular responses) and provide insight into the usefulness and appropriateness of utilizing different modes of exercise from both experimental and in-flight testing (i.e., both experimental and exercise countermeasure). The following Specific Aims are to be examined: Aim#1 [Characterize the Effect of Exercise Mode (Arm versus Leg cycling) on Exercise-Thermoregulatory Function and Cardiovascular Responses] Progress: Two subjects completed all protocols associated with this specific aim. From 30 minutes to the end of 60 minutes of exercise, the increase in internal temperature observed during Arm cycling was greater than during Leg cycling. While blood pressure responses to Arm and Leg cycling were not different, the heart rate responses to Arm exercise were greater from 30 minutes to the end of exercise compared to Leg cycling. Aim#2 (Characterize the Test, Re-test Reproducibility of Arm and Leg Cycling on Exercise-Thermoregulatory and Cardiovascular Responses) Progress: Eight subjects completed the protocols associated with this specific aim. In general, there were no differences between the body temperature and cardiovascular responses to Arm and Leg exercise between repeated visits (i.e., Arm and Leg exercises performed on repeated days). Presentations: In April of 2012, a poster presentation titled “Comparison of internal temperature responses to prolonged arm and leg ergometer exercise” was presented at the annual Experimental Biology meeting in San Diego, California. The abstract associated with that presentation was published in the FASEB Journal (March 2012).
Bibliography Type:	Description: (Last Updated: 03/13/2013)
Abstracts for Journals and Proceedings	Keller DM, Samels MR, Butts CL. "Comparison of internal temperature responses to prolonged arm and leg ergometer exercise." Experimental Biology 2012, San Diego, CA, April 21-25, 2012. Temperature Regulation and Fluid Balance session. FASEB Journal 2012 Apr;26(Meeting Abstract Supplement):1084.12. Search: http://www.fasebj.org/content/vol26/1_MeetingAbstracts , Apr-2012